



The Simultaneous Ascending Auction:

Rules and procedures

The simultaneous ascending auction was pioneered in 1994 by the U.S. Federal Communications Commission for the sale of airwave licenses. A simplified version was adopted by the British government in 2000 for the UK 3G auction. The objective of the auction was to distribute multiple licenses across multiple buyers each of whom desires to win exactly one license (in contrast, the FCC auction. The following is a summary of the most important rules of this auction type.

- 1) All objects are auctioned off through a single first price auction.
- 2) Bidding proceeds in multiple rounds, with the end of the auction determined by an ending rule.
- 3) In each round, potential buyers simultaneously and publicly submit their bids on a **single license only**.
- 4) New bids on a license must improve upon the current high bid (reservation price) by a *minimum increment*. The required increment is announced publicly each round and decreases in later rounds.
- 5) Bids are binding.
- 6) The provisionally winning bid on a license in a round is the highest bid that was submitted for that license in that round. It is automatically carried forward into the next round; all other bids expire at the end of the round.
- 7) Carried over provisionally winning bids may be renewed at the same bid level. The provisional winner is not allowed to switch which license being bid on.
- 8) Bidders that are not provisionally winning at the beginning of a round are free to bid on any license they so choose.
- 9) The “use-it-or lose-it” rule: You must place a bid in each round or dropout of the auction altogether. Once you dropout, you are no longer eligible to bid in future rounds.
- 10) The auction ends when all bidders that are not provisional winners have dropped out. Provisional winners win the license at the price of their bid.

This note was prepared as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.